

Chap 3

Inverse Functions

• $y = f(x)$ has an inverse function if and only if $y = f(x)$ is a one-to-one correspondence.

• f : a differentiable function with inverse f^{-1} and $f'(f^{-1}(a)) \neq 0$
 $\Rightarrow$ The inverse function is differentiable at a and

$$(f^{-1})'(a) = \frac{1}{f'((f^{-1})(a))}.$$

* Differentiation Rules of Logarithmic and Exponential Functions

$$(1) \ y = a^x \Rightarrow y' = a^x \ln a$$

$$(2) \ y = e^x \Rightarrow y' = e^x$$

$$(3) \ y = \log_a x \Rightarrow y' = \frac{1}{x \ln a}$$

$$(4) \ y = \ln |x| \Rightarrow y' = \frac{1}{x}$$

• If n is any real number and $f(x) = x^n$, then

$$f'(x) = nx^{n-1}$$

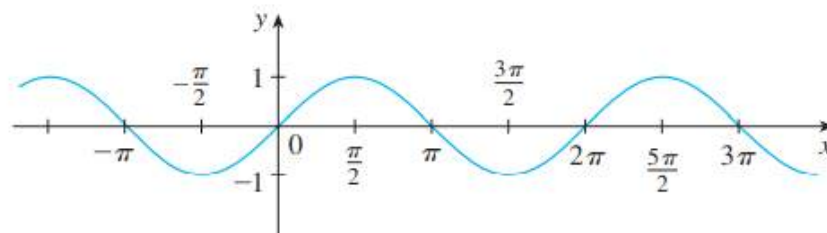
(Exam) Differentiate $y = x^{\sqrt{x}}$.

(sol)

§5 Inverse Trigonometric Functions

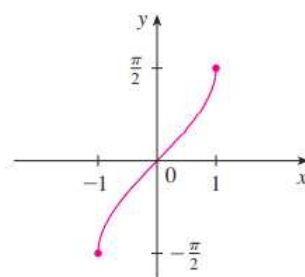
* Inverse Trigonometric Functions

$$(1) \ y = \sin x, \ -\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$$



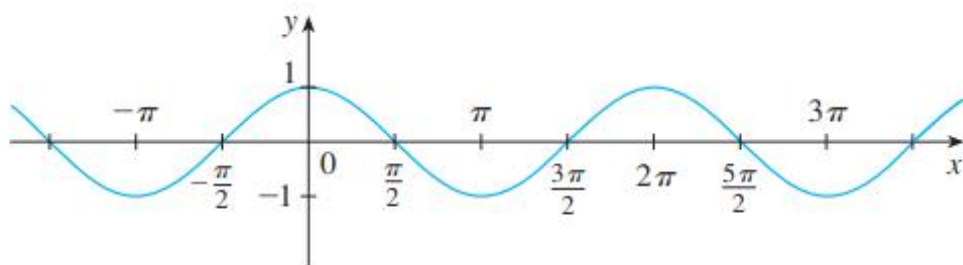
$$\xrightarrow{\text{inverse}} \ x = \sin y, \ -\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$$

$\Leftrightarrow y = \sin^{-1} x$ or $y = \arcsin x$: the inverse sine function



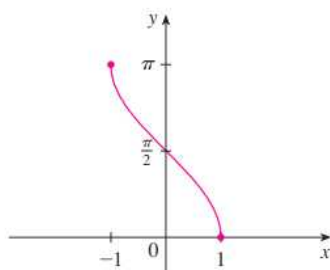
- Domain: $\{ x \mid -1 \leq x \leq 1 \}$
- Range: $\left\{ y \mid -\frac{\pi}{2} \leq y \leq \frac{\pi}{2} \right\}$
- Odd function ($\sin^{-1}(-x) = -\sin^{-1}x$)
- $\sin^{-1}(\sin x) = x$ for $-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$
- $\sin(\sin^{-1}x) = x$

(2) $y = \cos x, 0 \leq x \leq \pi$



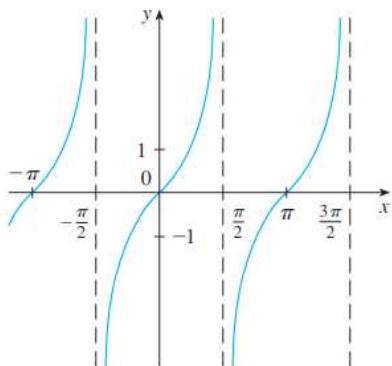
inverse $\rightarrow x = \cos y, 0 \leq y \leq \pi$

$\Leftrightarrow y = \cos^{-1} x$ or $y = \arccos x$: the inverse cosine function



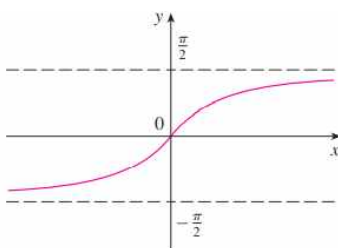
- Domain: $\{ x \mid -1 \leq x \leq 1 \}$
- Range: $\{ y \mid 0 \leq y \leq \pi \}$
- $\cos^{-1} x = \frac{\pi}{2} - \sin^{-1} x$

$$(3) \ y = \tan x, \ -\frac{\pi}{2} < x < \frac{\pi}{2}$$



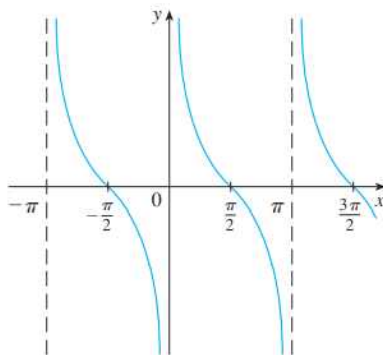
$$\xrightarrow{\text{inverse}} \ x = \tan y, \ -\frac{\pi}{2} < y < \frac{\pi}{2}$$

$\Leftrightarrow \ y = \tan^{-1} x$ or $y = \arctan x$: the inverse tangent function



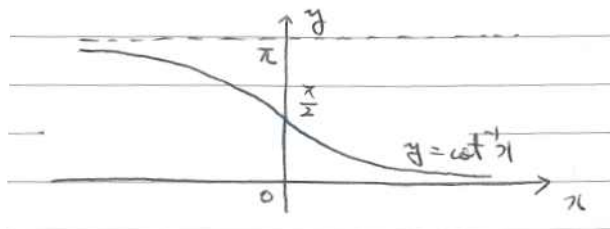
- Domain: $\mathbb{R} = \{ x \mid -\infty < x < \infty \}$
- Range: $\left\{ y \mid -\frac{\pi}{2} < y < \frac{\pi}{2} \right\}$
- Odd function ($\tan^{-1}(-x) = -\tan^{-1}x$)
- Asymptotes : $y = \pm \frac{\pi}{2}$
- $\lim_{x \rightarrow \infty} \tan^{-1} x = \frac{\pi}{2}$ and $\lim_{x \rightarrow -\infty} \tan^{-1} x = -\frac{\pi}{2}$

(4) $y = \cot x$, $0 < x < \pi$



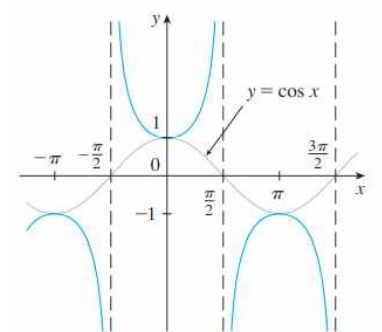
inverse $\rightarrow x = \cot y$, $0 < y < \pi$

$\Leftrightarrow y = \cot^{-1} x$ or $y = \operatorname{arccot} x$: the inverse cotangent function



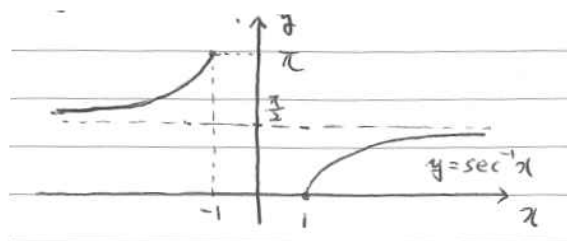
- Domain: $\mathbb{R} = \{ x \mid -\infty < x < \infty \}$
- Range: $\{ y \mid 0 < y < \pi \}$
- Asymptotes : $y = 0$ (the x -axis) and $y = \pi$
- $\cot^{-1} x = \frac{\pi}{2} - \tan^{-1} x$

(5) $y = \sec x$, $0 \leq x < \frac{\pi}{2}$, $\frac{\pi}{2} < x \leq \pi$



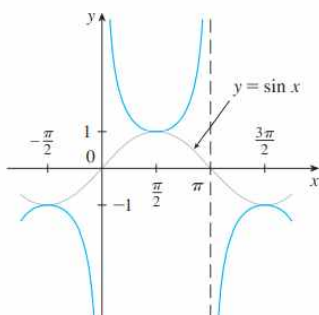
inverse $\rightarrow x = \sec y$, $0 \leq y < \frac{\pi}{2}$, $\frac{\pi}{2} < y \leq \pi$

$\Leftrightarrow y = \sec^{-1} x$ or $y = \operatorname{arcsec} x$: the inverse secant function



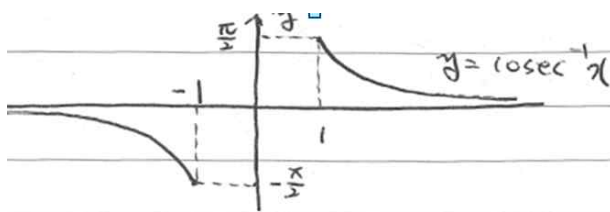
- Domain: $\{ x \mid x \leq -1, \ x \geq 1 \}$
- Range: $\left\{ y \mid 0 \leq y < \frac{\pi}{2}, \ \frac{\pi}{2} < y \leq \pi \right\}$
- Asymptotes : $y = \frac{\pi}{2}$

$$(6) \ y = \csc x, \quad -\frac{\pi}{2} \leq x < 0, \quad 0 < x \leq \frac{\pi}{2}$$



$$\xrightarrow{\text{inverse}} \quad x = \csc y, \quad -\frac{\pi}{2} \leq y < 0, \quad 0 < y \leq \frac{\pi}{2}$$

$\Leftrightarrow \quad y = \csc^{-1} x \quad \text{or} \quad y = \operatorname{arccsc} x$: the inverse cosecant function



- Domain: $\{ x \mid x \leq -1, \quad x \geq 1 \}$
- Range: $\left\{ y \mid -\frac{\pi}{2} \leq y < 0, \quad 0 < y \leq \frac{\pi}{2} \right\}$
- Odd function
- Asymptotes : $y = 0$ (the x -axis)
- $\csc^{-1} x = \frac{\pi}{2} - \sec^{-1} x$

* Derivatives of Inverse Trigonometric Functions

$$(1) \quad y = \sin^{-1}x \Leftrightarrow x = \sin y, \quad -\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$$

$$\Rightarrow 1 = \cos y y'$$

$$\begin{aligned} \Rightarrow y' &= \frac{1}{\cos y} \\ &= \frac{1}{\sqrt{1 - \sin^2 y}} \quad \left(\because \cos y > 0 \text{ for } -\frac{\pi}{2} < y < \frac{\pi}{2} \right) \\ &= \frac{1}{\sqrt{1 - x^2}} \end{aligned}$$

$$(2) \quad y = \cos^{-1}x = \frac{\pi}{2} - \sin^{-1}x$$

$$\Rightarrow y' = -\frac{1}{\sqrt{1 - x^2}}$$

$$(3) \quad y = \tan^{-1}x \Leftrightarrow x = \tan y, \quad -\frac{\pi}{2} < y < \frac{\pi}{2}$$

$$\Rightarrow 1 = \sec^2 y y'$$

$$\begin{aligned} \Rightarrow y' &= \frac{1}{\sec^2 y} \\ &= \frac{1}{1 + \tan^2 y} \\ &= \frac{1}{1 + x^2} \end{aligned}$$

$$(4) \quad y = \cot^{-1}x = \frac{\pi}{2} - \tan^{-1}x$$

$$\Rightarrow y' = -\frac{1}{1 + x^2}$$

$$(5) \quad y = \sec^{-1}x = \cos^{-1}\left(\frac{1}{x}\right)$$

$$\Rightarrow y' = -\frac{1}{\sqrt{1-\left(\frac{1}{x}\right)^2}}\left(-\frac{1}{x^2}\right)$$

$$= \frac{1}{x^2 \sqrt{1-\frac{1}{x^2}}}$$

$$= \frac{1}{x^2 \sqrt{\frac{x^2-1}{x^2}}}$$

$$= \frac{|x|}{|x|^2 \sqrt{x^2-1}}$$

$$= \frac{1}{|x| \sqrt{x^2-1}}$$

$$(6) \quad y = \csc^{-1}x = \frac{\pi}{2} - \sec^{-1}x$$

$$\Rightarrow y' = -\frac{1}{|x| \sqrt{x^2-1}}$$

Exam 1 Evaluate (a) $\sin^{-1}\left(\frac{1}{2}\right)$ and (b) $\tan\left(\arcsin\frac{1}{3}\right)$.

(sol)

Exam 2 If $f(x) = \sin^{-1}(x^2 - 1)$, find (a) the domain of f , (b) $f'(x)$, and (c) the domain of f' .

(sol)

Exam 3 Simplify the expression $\cos(\tan^{-1}x)$.

(sol)

Exam 4 Evaluate $\lim_{x \rightarrow 2^+} \arctan\left(\frac{1}{x-2}\right)$.

(sol)

Exam 5 Differentiate $f(x) = x \tan^{-1} \sqrt{x}$.

(sol)

(Exam) Differentiate (a) $y = x \sin^{-1} x + \sqrt{1-x^2}$ and

(b) $y = \cos^{-1} \left(\frac{b+a \cos x}{a+b \cos x} \right)$, $0 < x < \pi$, $a > b > 0$.

(sol)

(cf) $\int \sin^{-1} x \, dx = x \sin^{-1} x + \sqrt{1-x^2} + C$

(Exam) Find the value of $\sin^{-1}\left(\frac{3}{5}\right) + \sin^{-1}\left(\frac{4}{5}\right)$.

(sol)

(Exam) Show that $\sec^{-1}a = \cos^{-1}\frac{1}{a}$.

(sol)

(cf) (i) $\csc^{-1}a = \sin^{-1}\frac{1}{a}$

(ii) $\cot^{-1}a = \begin{cases} \tan^{-1}\frac{1}{a} & (a > 0) \\ \pi + \tan^{-1}\frac{1}{a} & (a < 0) \end{cases}$